

SY B.Tech. – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PRACTICAL ASSIGNMENT- 2

1. Student Details

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Division	A
Batch	A-2
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Date of Submission	09-09-2026

2. Practical Details

Practical No.	2
Practical Title	Class, Constructor, Destructor and User-Defined Functions
Date Performed	09-09-2026
Faculty Name	Dr. Khushal Khairnar

3. Problem Statement

To implement various types of constructors and a member function in C++ using a Car class.

4. Objectives

- 1.To understand different types of constructors.
- 2.To implement default, parameterized, and copy constructors.
- 3.To implement a member function to display car details.

5. Expected Learning Outcomes

- Understand the concept of constructors.
- Implement different types of constructors in C++.
- Use member functions with class objects.

6. Theory / Concept

A constructor is a special member function that is automatically called when an object is created. In this program, three types of constructors are used: default, parameterized, and copy constructor. The display() function is a member function used to display the car details...

7. Algorithm / Pseudocode

START

Step 1: Define a Car class with brand, model year, and price.

Step 2: Define a default constructor to initialize car details.

Step 3: Define a parameterized constructor to initialize values passed as arguments.

Step 4: Define a copy constructor to copy details from one object to another.

Step 5: Define the display() member function to display car details.

Step 6: Create objects using all three constructors.

Step 7: Call display() for each object.

08. Technology / Tools / Libraries Used

Technology / Tool C++	Purpose
IDE / Platform Dev-	Programming Language
C++ / Code::Blocks	IDE / Platform
iostream	Input and Output

09. Implementation / GitHub Repository

Particular	Details
GitHub Repository URL	asfd
Repository Name	Assigenment2
Branch Used	main 1
Commit ID / Latest Commit	Add file via upload
Particular	Details
Date of Last Commit	09-09-2026

10. Test Cases / Input and Output

Test Case	Input	Expected Output	Actual Output	Status
1	Car c1;	Toyota, 2022, 1500000 displayed	Toyota, 2022, 1500000 displayed	Pass
2	Car c2("Hyundai", 2024, 1450000);	Hyundai, 2024, 1450000 displayed	Hyundai, 2024, 1450000 displayed	Pass

11. Results / Observations

The program was successfully executed and demonstrated the use of three types of constructors: default, parameterized, and copy constructor. The display() member function was used to display the car details such as brand, model year, and price. The output was obtained correctly for all the created objects.

12. Conclusion

Thus, the program successfully implemented and demonstrated default, parameterized, and copy constructors in C++. It also helped in understanding how a member function can be used with objects to display their data. The practical provided a basic understanding of constructor overloading and object initialization in C++.

13. Student Declaration

I hereby declare that the above practical work has been completed by me and that the implementation, analysis, and answers are my own work.

Student Name	PRN	Signature
Homesh Sawarkar	202501110069	Homesh A.S